world picture of quantum mechanics as in that of classical physics. The difference consists only in the symbols used and in the mathematics applied. Accordingly, in the realm of quantum mechanics, just as formerly in classical physics, the uncertainty in the prediction of the occurrences of the sense world is reduced to an uncertainty in the correlation of world picture and sense world, in other words, to an uncertainty in the translation of the symbols of the world picture into the sense world, and *vice versa*. The fact that this double uncertainty is involved is the most impressive proof of the importance of retaining the principle of determinism within the world picture.

Nevertheless, even a cursory glance shows how far the world picture of quantum mechanics has shifted from the sense world, and how much more difficult it is to translate an occurrence from the world picture of quantum mechanics into the sense world or *vice versa*, than was the case in classical physics. In the latter, the meaning of every symbol was immediately and directly intelligible; the position, the velocity, the momentum, the energy of a material point could be determined more or less accurately by measurements, and there appeared no obvious reason to doubt that a steady improvement of the technique and methods of measurement could not eventually reduce the remaining factor of uncertainty below any desired margin. On the other hand, the wave function of quantum mechanics supplies no direct clue whatever for any obvious interpretation of the sense world, simply because it does not refer to ordinary space at all but rather to the configurational space which has as many dimensions as there are coordinates present in the physical structure under consideration. Moreover—and this is where the real trouble lies—the wave function does not give us the values of the coordinates as functions of the time, but merely the probability that the coordinates may possess certain specific values at some specified time.

The indeterminists again seized on this circumstance as the occasion for a new attack against the law of causality. And this time their attack actually seems to promise them success; for on the basis of measurements it is possible to assign merely a statistical significance to the wave function. Nonetheless, once more the defenders of strict causality have the same way out as before: The assumption that the question concerning the meaning of a certain symbol of the world picture of quantum mechanics, such as a material wave, has no definite sense unless one also specifies the condition of the particular measuring apparatus used for translating the symbol into the sense world. For this reason, one is led to talk about the causal effect of the measuring instrument that is employed; and one is thereby expressing the fact that the particular uncertainty in question is determined at least in part by the circumstance that the magnitude of the value to be measured is dependent in a certain regular manner on the method used for measuring it.

However, this auxiliary hypothesis shunts the entire question to a track, the